Assessment - Answer Key

Chapter 2: Diversity in the Living World

I. Choose the correct answer :

1. b) Lotus survives in water because its **waxy leaves float and do not get waterlogged.**
2. d) Correct answer is **all of these**, because camel, yak, and whale live in different habitats.
3. **b) Maize** is a monocot plant with one cotyledon.
4. b) A polluted pond with few fish shows **loss of biodiversity**.
5. c) Amphibians breathe through **lungs and moist skin**, which help them survive both on land and water.
6. d) Correct answer is **all of these**, as different animals use body parts for movement.

II. Match the following :

7. Monocot seed – One cotyledon
8. Dicot seed – Two cotyledons
9. Rhododendron – Mountain plant
10. Whale – Aquatic mammal
11. Frog – Amphibian

III. Answer the given questions as true or false:

12. False – Most fibrous-root plants have parallel venation, but exceptions exist.
13. False – Camels do not sweat; they conserve water.
14. False – Amphibians live on land and water.
15. True – Sacred groves protect forests and biodiversity.
16. True – Seeds help classify plants.

IV. Differentiate between the following:

17. Hot vs Cold Desert Camel

- Hot desert: One hump, survives with little water, short hair.
- Cold desert: Two humps, thick hair, adapted to cold.

18. Herbs, Shrubs, Trees

- Herbs: Small, soft stems, e.g., mint.
- Shrubs: Medium, woody stems, e.g., rose.
- Trees: Tall, woody trunk, e.g., neem.

V. Answer the following questions in brief:

19. Rhododendrons have small leaves to reduce water loss and protect from cold.
20. Biodiversity is not the same everywhere because climate and resources differ, like deserts have fewer species while rainforests have more.
21. Frogs have lungs and moist skin for breathing and webbed feet for swimming, making them fit for both water and land.
22. Sacred groves are patches of forests protected by people for cultural reasons, helping biodiversity survive.

VI. Answer the following questions in detail:

23. Classifying Plants in a School Garden

If I were a researcher classifying plants in a school garden, I would use features like height, venation, and roots. First, I would divide them into herbs, shrubs, and trees based on their stem and height. Then I would check their leaves: if the veins are parallel, the plant is a monocot, and if the veins are net-like, it is a dicot. Finally, I would examine their roots: fibrous roots belong to monocots, while taproots belong to dicots. This method would give a clear, systematic record of the plants and make the study easier.

24. **Aquatic vs Terrestrial Habitats**

Aquatic habitats are water-based, and plants and animals here adapt to survive in water. For example, lotus plants have waxy leaves that float, and fish have gills for breathing. Terrestrial habitats are land-based, where organisms adjust to hot, cold, or dry conditions. Cactus has spines instead of leaves, and camels have humps to store fat for survival in deserts. Both habitats present challenges, and organisms evolve different features to survive. This comparison shows the diversity of life forms in different environments.

25. **Grouping and Biodiversity**

Grouping plants and animals helps us understand biodiversity in a clear way. If we randomly study each organism, it becomes confusing, but grouping shows patterns and similarities. For example, grouping plants as monocots and dicots links seeds, roots, and leaves together. Grouping animals by movement, like flying, swimming, or crawling, shows how their bodies are adapted. By grouping, scientists can classify endangered species and plan conservation efforts. It also helps students and communities to appreciate the richness of biodiversity and the need to protect it.

Mrithurica.

Assessment Question

Chapter 2: Diversity in the Living World

Marks: 40 **Duration: 1 hr 15 mins**

I. Choose the correct answer: **(6 × 1 = 6)**

1. Which adaptation helps lotus survive in water?
   a) Thick bark b) Waxy leaves c) Fibrous roots d) None
2. Which is a correct animal-habitat pair?
   a) Camel – desert b) Yak – mountains c) Whale – ocean d) All of these
3. Which one is a monocot plant?
   a) Pea b) Maize c) Neem d) Mustard
4. Which of the following shows reduction in biodiversity?
   a) Forest with many species b) Polluted pond with few fish
   c) Farm with mixed crops d) Mountain with many herbs
5. Which feature helps amphibians survive?
   a) Gills only b) Lungs only c) Both lungs and moist skin d) None
6. Which movement is correctly matched?
   a) Fish – fins b) Bird – wings c) Human – legs d) All of these

II. Match the following: **(5 × 1 = 5)**

7. Monocot seed - Amphibian
8. Dicot seed - Aquatic mammal
9. Rhododendron - One cotyledon
10. Whale - Two cotyledons
11. Frog - Mountain plant

III. Answer the given questions as true or false: **(5 × 1 = 5)**

12. All plants with fibrous roots show parallel venation.
13. Camels sweat a lot to survive deserts.

14. Amphibians can only survive in water.
15. Sacred groves are cultural methods of biodiversity protection.
16. Seed structure can help classify plants.

IV. Differentiate between the following: (2 × 3 = 6)

17. Hot desert camel and Cold desert camel.
18. Herbs, Shrubs, and Trees.

V. Answer the following questions in brief: (4 × 2 = 8)

19. Why do rhododendrons at high mountains have small leaves?
20. Why is biodiversity not the same in desert and rainforest?
21. How are frogs adapted to both land and water?
22. Why should we conserve community-protected forests?

VI. Answer the following questions in detail: (Any two) → document (2 × 5 = 10)

23. Imagine you are a researcher classifying plants in your school garden. Describe the criteria you will use and why.
24. Compare aquatic and terrestrial habitats with examples, adaptations, and challenges.
25. Evaluate how grouping plants and animals helps us understand biodiversity and prevents its loss.

BANNARI AMMAN PUBLIC SCHOOL

Additional Sheet No. 1

Diversity in the living world

Name: Mrithunica. A

Grade: V

Date: 14/10/2025

Subject: Science

31/40

A Amarkarthik

I Choose the correct answer

1 Fibrous roots

d 2 All of these

b 3 Maize

b 4 Polluted pond with few fish

d 5. None

d 6 All of these

II Match the following

7 Monocot seed - One cotyledon

8 Dicot seed - Two cotyledons

5 9 Rhododendron - Mountain plant

10 Aquatic mammal - Whale

11 Frog - Amphibian

III Answer the given question as true or false

12 True

13 True

3 14 False

15 True

16 True

12

IV Differentiate between the following

17) Hot dessert camel, Cold dessert camel

Hot dessert camel	Cold dessert camel
1 hump	2 hump
Thick eye lashes	Normal eye lashes
Thin fur	Thick fur
Long legs	Short legs

3

18)

Herbs	Shrubs	Trees
Soft, green stems	thin wood stems	Thick wood trunk
small	Medium	large
Medicinal value	some medicinal value	not much medicinal value

2½

V Answer the questions in brief

19) Rhododendrons:
They have small leaves because of the strong winds. If wind blows and they might get cut or fall but even if they fall there will still be a lot left. If there is snow fall then the snow can easily fall and not bunch up at the top causing the tree to collapse.

1

20) Biodiversity
Biodiversity is not same in forests and deserts because there are many different animals and plants, for example take a cactus and a mango tree. They both are adapted to different conditions like in the desert it is hot so cactuses have spines instead of leaves because leaves lose too much water through their stomatas but mango trees have leaves and adapted to the normal conditions. Same goes for animals. Camels are adapted to the hot

2

8½

desert. But other animals in the forests are not and are adapted to normal conditions. This is how biodiversity differs

21) frogs

frogs are adapted to both land and water because they are amphibians which means they live part of their lives in water and part in land. Because of this they are adapted to this and can swim well and also have lungs gills to breath underwater.

22) Community protected forests

We should conserve community protected forests because they are vital to biodiversity. There are a lot of unique plants and animals there and they will lose their home if we dont conserve them. they too play vital role on Earth in converting carbon dioxide into oxygen

2

VI Answer the following questions in detail

23) Classification of plants

I would classify plants wether they are herbs, shrubs or trees because then I can learn more things about them in the garden which would be useful to learn about their medicinal value how does it differ from herb shrub and tree. I would learn about their size and stems which would be useful for differentiating them. I would also learn about their differences similarities and how unique they are, just because of this method of grouping.

3

24) aquatic and terrestrial habitats

aquatic habitats and terrestrial habitats are different. They come with unique adaptations and challenges. some of them are

6

that they both have predators and prey but have different issues like different kinds of pollution and get affected all because of us humans. In aquatic habitats there is water pollution. In terrestrial habitats there is air pollution, climate change and deforestation, these animals have different adaptations like in the water fish have gills to
A½ breath but mammals on land have lungs to breath. Underwater fish have streamlined bodies to move swiftly while mammals on land have special hooves and paws some to move swiftly and nimbly some to run fast.

4½